

● MEDIUM

● **ADVANCED MATH**

Advanced Math

05

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1

ADVANCED MATH

$$52x^3 + 64x^4 - 81 = 0$$

How many distinct real solutions does the given equation have?

- A. Exactly two
- B. Exactly three
- C. Exactly five
- D. Exactly seven

Which of the following is equivalent to the sum of $3x^4 + 2x^3$ and $4x^4 + 7x^3$?

- A. $16x^{14}$
- B. $7x^8 + 9x^6$
- C. $12x^4 + 14x^3$
- D. $7x^4 + 9x^3$

x	gx
-1	25
0	1
1	$\frac{1}{25}$
2	$\frac{1}{625}$

For the exponential function g , the table shows four values of x and their corresponding values of gx . Which equation defines g ?

A. $gx = -25^x$

B. $gx = -\frac{1}{25}^x$

C. $gx = 25^x$

D. $gx = \frac{1}{25}^x$

$$x = 8ab + 9$$

The given equation relates the positive numbers a , b , and x . Which equation correctly expresses a in terms of b and x ?

A. $a = \frac{x}{8} - b + 9$

B. $a = \frac{x}{8b + 9}$

C. $a = \frac{8b + 9}{x}$

D. $a = 8xb + 9$

The expression $\frac{24}{6x + 42}$ is equivalent to $\frac{4}{x + b}$, where b is a constant and $x > 0$. What is the value of b ?

- A. 7
- B. 10
- C. 24
- D. 252

A model predicts that the population of Bergen was 15,000 in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$?

- A. $p = 0.9615,000^x$
- B. $p = 1.0415,000^x$
- C. $p = 15,0000.96^x$
- D. $p = 15,0001.04^x$

Q7

GRID-IN

ADVANCED MATH

In the xy -plane, what is the y -coordinate of the point of intersection of the graphs of $y = (x-1)^2$ and $y = 2x-3$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$(4x^3 - 5x^2 + 3) - (6x^3 + 2x^2 - x)$$

Which of the following expressions is equivalent to the expression above?

- A. $-10x^3 - 3x^2 + x + 3$
- B. $-2x^3 - 7x^2 + x + 3$
- C. $-2x^3 - 3x^2 + x + 3$
- D. $10x^3 - 7x^2 - x + 3$

The function $f(t) = 40,000 \cdot 2^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

- A. 2
- B. 790
- C. 1,580
- D. 40,000

$$x^2 - 12x + 27 = 0$$

How many distinct real solutions does the given equation have?

- A. Exactly two
- B. Exactly one
- C. Zero
- D. Infinitely many